

Hao (Evie) Wen

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EDUCATION

California Polytechnic State University

B.S., Computer Engineering

- GPA: 3.6 / 4.0

September, 2022 – June, 2026

San Luis Obispo, CA

TECHNICAL SKILLS

- **Languages:** C, Python, Java, SystemVerilog, Assembly, RISC-V
- **Hardware:** Oscilloscopes, Signal Generators, Multimeters, Soldering, Breadboard Development, FPGA Boards, STM32 Microcontrollers, Arduino Microcontrollers
- **Software:** LTSpice, MATLAB, Xilinx Vivado, Git / GitHub, Linux, VSCode, KiCad

PROJECTS

- **Smart Electric Vehicle Charging System** | *Python / Serial Communications / Raspberry Pi*
 - Developed custom OCPP v1.6-compliant server deployed on Raspberry Pi for EV charging control
 - Integrated IPEM energy-monitoring board to read CT clamp currents over serial communication
 - Implemented FSM control loop enabling dynamic charge rate adjustment based on power draw
 - Tested and verified system compatibility with commercially viable Wallbox Pulsar Plus EVSE
 - Collaborated in multidisciplinary team, gaining experience in project scheduling, research and development process, and overall systems integration
- **STM32 Autonomous Vehicle** | *STM32 / Circuits / C*
 - Designed and implemented autonomous vehicle using STM32 Nucleo-L476RG, DC motors, and servo-mounted ultrasonic sensor for obstacle detection
 - Implemented “always-take-left” pathfinding with bi-directional scanning and reversing until a clear path is found
 - Enabled full mobility using 9V battery-powered operation
- **Multi-Cycle RISC-V Microprocessor** | *RISC-V / FPGA / Vivado / SystemVerilog*
 - Enhanced prior single-cycle RISC-V microprocessor by implementing 5-stage pipeline and 8-set, 2-way associative cache, improving instruction throughput
 - Designed pipeline control to handle data/control hazards using forwarding and static branch prediction
 - Implemented write-back cache policy with 1-bit LRU replacement, deployed and tested on BASYS-3 FPGA using AMD Xilinx
- **Distortion Guitar Pedal** | *Circuits / LTSpice / KiCad*
 - Designed and tested custom guitar pedal circuit using input of real physical electric guitar instruments to output modified and distorted signal
 - Deployed using analog circuit electronics including op-amps, soldered input / output jacks, adjustable potentiometers on electrical breadboard
 - Ensured functionality with actual on-the-market guitar amplifiers and traditional 9V AC power adapters
 - In the process of transferring breadboard circuit design onto an electrical PCB with soldered components using KiCad
- **Frequency-Actuated Solenoid** | *LTSpice*
 - Designed and implemented analog frequency-actuated solenoid system using ceramic microphone input
 - Processed audio signal through BJT common-collector amplifier and IC comparator to detect user-defined frequency threshold
 - Controlled solenoid actuation via PMOS switching, toggling between V_{CC} and GND at frequency threshold